

# IoT - Based Smart Industrial Monitoring and Predictive Maintenance System Using ESP32

**Prepared By: Aswath Karthick Sivasankar**

## **Project Overview :**

This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Project Overview for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation.

## **Problem Statement :**

This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Problem Statement for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation.

## Objectives :

This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Objectives for the IoT-Based Smart Industrial

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## System Architecture :

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## Hardware Components :

This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation. This section explains Hardware Components for the IoT-Based Smart Industrial Monitoring and Predictive Maintenance System using ESP32. The project focuses on real-time monitoring, predictive maintenance, cloud connectivity, reliability improvement and industrial automation.

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## Software Requirements :

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### **Working Principle :**

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## **ESP32 Code Logic :**

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### Result :

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## **Applications :**

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## Future Enhancements :

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## **Conclusion :**

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